Bayesian Networks I – Representation and Exact Inference – graphical structure, conditional independence, CPTs, joint distributions, inference by enumeration / variable elimination

CS 4880: Artificial Intelligence

Lecture 19 – October 9, 2026

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Today

- Course unit: Course sequence
- Big idea: Bayesian Networks I – Representation and Exact Inference – graphical structure, conditional independence, CPTs, joint distributions, inference by enumeration / variable elimination
- Placeholder for announcements and context.

Learning Goals

- Define the key vocabulary for this lecture.
- Work through one representative example.
- Connect today's idea to the previous and next class meetings.

Core Ideas

Placeholder

Replace these bullets with the final lecture material, examples, and in-class activities.

- Main concept 1.
- Main concept 2.
- Main concept 3.

Example or Activity

In class

Add a short worked example, code trace, proof sketch, discussion prompt, or board activity here.

Wrap-Up

- What should students be able to do after today?
- What should they practice before the next meeting?
- What question should they bring to class next time?